

TABLE I: Classification of Possible Eavesdropper Scenarios in Anti-Drone Physical Layer Security

Category	Eve Scenario	Description	Possible Identification Method
Mobility-Based	Static Eve	Eve remains fixed at a particular position and observes the communication link between Alice and Bob. This type of Eve has almost constant distance and channel behavior.	RSSI analysis, CSI monitoring, location verification, secrecy capacity observation.
	Hovering Eve	Eve behaves like a drone that stays nearly fixed in air. Although it appears stationary, small hovering movements can still create minor channel variations.	Small-scale channel variation analysis, Doppler shift observation, angle of arrival estimation.
	Dynamic Eve	Eve moves continuously with time. Its position, received power, Doppler profile, and channel gain change according to its movement.	Mobility tracking, Doppler analysis, time-varying CSI, trajectory verification.
Activity-Based	Passive Eve	Eve only listens to the confidential signal and does not transmit anything. This is one of the most difficult cases because Eve may not leave any direct signal evidence.	Indirect detection using secrecy capacity, outage probability, BER, SNR, and suspicious channel behavior.
	Active Eve	Eve transmits signals into the network. It may send fake pilots, false responses, or unauthorized access requests.	Pilot authentication, RF fingerprinting, CSI mismatch detection, timing analysis.
	Jamming Eve	Eve intentionally transmits interference to reduce the received signal quality at Bob and disturb the legitimate communication link.	Sudden SNR degradation, BER increase, interference power detection, spectrum monitoring.
	Transmitting and Receiving Eve	Eve listens to Alice and also transmits signals toward Alice or Bob. It may disturb, mislead, or try to enter the communication link.	Bidirectional signal monitoring, abnormal packet detection, RF fingerprinting, traffic behavior analysis.
Identity-Based	Eve Acting as Bob	Eve tries to impersonate Bob and convince Alice that it is the legitimate receiver. This case is highly critical in command and control communication.	Channel-based authentication, location verification, RF fingerprinting, challenge-response verification.
	Relay Eve	Eve acts as an intermediate node by receiving the signal from Alice and forwarding or modifying it toward Bob. This resembles a man-in-the-middle behavior.	Latency analysis, relay path detection, timing mismatch, channel inconsistency checking.
	Unknown Drone	A detected drone has no confirmed identity. The system must decide whether it is a legitimate drone or an unauthorized drone.	Registered ID checking, channel profile matching, location confirmation, authentication verification.
	Compromised Bob	A legitimate receiver is captured, misused, or controlled by an attacker and starts behaving abnormally.	Behavior monitoring, abnormal request detection, trust score evaluation, historical profile comparison.
Number-Based	Single Eve	Only one unauthorized drone or receiver is present and tries to listen to or attack the communication link.	Single-node channel analysis, secrecy capacity estimation, received power comparison.
	Multiple Independent Eves	More than one Eve is present, but each attacker operates independently without sharing information with others.	Multi-node localization, angle of arrival estimation, distributed signal monitoring.
	Colluding Eves	Multiple Eves cooperate and share their received information to improve the chance of decoding Alice's confidential message.	Cooperative attack detection, correlation analysis, multi-receiver secrecy analysis.
Special Cases	Only One Drone Present	If only one drone is detected, the system cannot directly assume that it is Bob. It must verify whether the drone is legitimate or suspicious.	Authentication, expected trajectory checking, RF fingerprint matching, CSI comparison.
	Intermittent Eve	Eve appears only during selected time intervals and remains silent at other times to avoid continuous detection.	Time-window analysis, anomaly detection, periodic channel monitoring.